

VAIAKU, Tuvalu

WMO: 916430

Lat: 8.522S

Lon: 179.198E

Elev: 2

StdP: 101.30

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 80705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.8	25.0	21.8	16.5	27.8	22.1	16.7	27.9	12.3	27.9	11.3	28.1	3.8	20	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	4.3	32.7	27.4	32.2	27.1	32.1	27.1	28.2	31.3	28.0	31.1	27.7	30.8	3.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.5	27.1	22.8	30.5	26.8	22.4	30.2	91.3	31.9	90.0	31.2	88.3	31.0	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.5	8.4	23.3	33.5	0.9	0.3	22.7	33.8	22.1	34.0	21.6	34.2	21.0	34.4		
			22.0	29.6	0.5	0.8	21.6	30.1	21.4	30.6	21.1	31.0	20.7	31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	28.7	28.6	28.7	29.0	29.1	29.0	28.9	28.7	28.9	28.9	29.0	28.9	
	DBStd	1.01	1.16	1.03	1.06	1.01	0.97	0.92	0.93	0.98	0.90	0.99	1.00	1.03	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6889	581	522	579	572	591	570	584	579	567	587	570	587	
	CDD18.3	3848	322	288	321	322	332	320	326	321	317	329	320	329	
	CDH23.3	47033	3953	3461	3889	3987	4102	3937	3923	3826	3875	4059	3946	4076	
	CDH26.7	18428	1546	1283	1483	1632	1658	1569	1485	1399	1508	1631	1592	1644	
Wind	WSAvg	3.8	3.9	4.2	3.4	3.3	3.2	4.2	4.5	4.2	4.2	3.3	3.2	3.5	
Precipitation	PrecAvg	3467	408	315	353	270	238	217	246	251	229	252	279	413	
	PrecMax	4833	891	750	978	620	615	417	564	670	680	483	661	837	
	PrecMin	2226	167	93	148	63	46	94	72	94	66	60	51	152	
	PrecStd	617	158	155	152	143	116	94	111	121	135	98	136	177	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	32.2	32.4	32.8	32.5	32.4	32.2	32.2	32.2	32.5	33.0	33.0	
		MCWB	27.5	27.1	27.2	27.6	27.5	27.4	27.3	27.2	27.2	27.3	27.5	27.7	
	2%	DB	32.1	31.9	32.0	32.1	32.1	32.0	31.9	31.8	32.0	32.1	32.2	32.2	
		MCWB	27.0	26.9	27.0	27.2	27.2	27.3	27.1	27.0	27.0	26.9	27.0	27.1	
	5%	DB	31.7	31.2	31.4	31.8	31.6	31.3	31.2	31.1	31.2	31.7	31.9	31.8	
		MCWB	26.9	26.7	26.8	27.1	27.1	27.1	26.9	26.9	26.8	26.8	26.8	27.0	
	10%	DB	31.1	30.9	31.0	31.1	31.1	31.0	30.8	30.8	30.9	31.1	31.2	31.1	
		MCWB	26.6	26.6	26.7	26.9	27.0	27.0	26.7	26.7	26.7	26.6	26.5	26.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	28.2	28.1	28.2	28.3	28.2	28.1	28.1	28.3	28.2	28.4	28.4
			MCDB	31.2	31.0	31.3	31.5	31.1	31.2	31.1	30.9	31.0	31.4	31.7	31.7
2%		WB	27.7	27.6	27.6	27.8	27.8	27.8	27.6	27.5	27.7	27.6	27.8	27.7	
		MCDB	30.7	30.5	30.9	31.1	30.9	30.7	30.6	30.5	30.6	31.0	31.2	31.0	
5%		WB	27.3	27.2	27.2	27.5	27.5	27.5	27.2	27.2	27.2	27.2	27.3	27.4	
		MCDB	30.4	30.1	30.4	30.7	30.5	30.4	30.2	30.2	30.3	30.6	30.6	30.6	
10%		WB	27.0	26.8	26.9	27.1	27.2	27.1	26.9	26.8	26.9	26.8	26.9	27.0	
		MCDB	30.1	29.7	30.0	30.3	30.2	30.0	29.8	29.9	30.0	30.1	30.1	30.2	
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.3	4.4	4.3	4.3	4.1	4.1	4.2	4.2	4.5	4.5	4.5	
		MCDBR	4.8	4.7	4.8	4.7	4.6	4.5	4.5	4.6	4.6	4.9	5.0	5.1	
		MCWBR	2.7	2.6	2.6	2.5	2.5	2.5	2.5	2.6	2.6	2.8	2.7	2.7	
	5% WB	MCDBR	4.5	4.4	4.5	4.5	4.3	4.2	4.2	4.4	4.3	4.6	4.8	4.7	
		MCWBR	2.8	2.7	2.6	2.6	2.6	2.6	2.5	2.6	2.7	2.8	2.8	2.7	
Clear-Sky Solar Irradiance	taub	0.412	0.413	0.394	0.390	0.396	0.412	0.414	0.418	0.417	0.405	0.406	0.405		
	taud	2.559	2.548	2.624	2.639	2.607	2.538	2.527	2.514	2.505	2.554	2.560	2.576		
	Ebn at Noon	929	924	927	904	869	838	844	865	892	920	928	933		
	Edh at Noon	108	109	99	94	92	96	99	105	110	107	107	106		
All-Sky Solar Radiation	RadAvg	5.35	5.32	5.65	5.46	4.89	4.62	4.78	5.29	5.93	6.01	5.68	5.50		
	RadStd	0.60	0.65	0.50	0.53	0.34	0.32	0.38	0.32	0.39	0.48	0.44	0.40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page